

COURSE DESCRIPTION			
Program	Wood and Paper Technology		
Course Title	Botany and Anatomical Structure		
Course Code	2121	Semester	II
Type of Course	Theory	Contact Hours	60
Teaching Scheme (L: T:P:R)	4:0:0:0	Credits	4
CIE Marks	40	ESE Marks	60

Introduction

Botany is the branch of biology that deals with the scientific study of plants. It includes the study of plant structure, classification, growth and economic importance. In wood and paper technology, botany is important for understanding the structure and composition of woody plants used as raw materials.

Plants provide cellulose, hemicellulose, and lignin which are essential components of wood and paper. Knowledge of plant anatomy and tissues such as xylem, fibers and vessels helps in identifying suitable timber species and improving pulp and paper production. Thus botany forms the basic scientific foundation for the wood and paper industry.

Course Objectives

1	To impart knowledge on the different types of classification in the plant kingdom.
2	To familiarize with the plants having immense economic importance.
3	To impart an insight into the internal structure and reproduction of plants.
4	To enable students to identify economically important tree species used in wood, pulp and paper industries.

Course Pre-requisite

Topic	Course code	Course Title	Semester
Basic knowledge in Botany			Secondary Education

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Summarize plant diversity.	Understand
CO2	Describe the plant taxonomy and morphology of plant parts.	Understand
CO3	Illustrate plant anatomy and plant tissues.	Understand
CO4	Discuss the plant body structure, wood anatomy and wood defects.	Understand

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	-	-	2	2	-	-	1	-	1
CO2	3	2	-	-	2	1	-	-	1	-	1
CO3	3	1	-	-	1	-	-	-	-	-	1
CO4	3	3	-	2	3	2	-	-	1	-	1
Total	12	8	-	2	8	5	-	-	3	-	4
Correlation Level	3	2.67	-	2	2	1.67	-	-	1	-	1

Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme						
							Theory			Practical			Total
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total	
4	0	0	0	6	60	4	40	60	100	0	0	0	100

L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

Syllabus - Major Topics

Module	Title	Major Topics	Instructional Hours	
			L	T
Module I	Plant diversity	Outline plant classification - Salient features and Economic importance of Algae, Fungi, Bryophytes, Pteridophytes and Gymnosperms	15	0
Module II	Plant morphology and Plant taxonomy	Root, Stem, Leaf and Flower morphology - Inflorescence - Significance of plant taxonomy - Angiosperm families	15	0
Module III	Plant anatomy	Structure and composition of plant cell - Characteristic features of different plant tissues - Theories of meristem - Vascular bundle	15	0
Module IV	Primary and Secondary structure, Wood anatomy and wood defects	Primary structure of stem, root and leaf - Secondary structure of stem and root - wood anatomy - wood defects	15	0

Detailed Syllabus

Subtopic	Student Learning Outcome	Mode of delivery (L/T)	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
Plant classification	Explain Two kingdom classification (Carolus Linnaeus, 1735) and general characters of each kingdom)	L	CO1	Remember	1
Plant classification	Explain Phylogenetic classification (August W Eichler, 1878)	L	CO1	Remember	1
General account on Fungus	Explain Salient features and economic importance of fungus with examples	L	CO1	Understand	1.5
General account on Fungus	Identify types of wood defects caused by Fungus- Blue stain, Brown rot, Dry rot, Heart rot, Sap stain, Wet rot, White rot	L	CO1	Understand	1.5
General account on Algae, Bryophytes and Pteridophytes	Explain Salient features and Economic importance of Algae, Bryophytes and Pteridophytes with examples	L	CO1	Understand	5
General account on Gymnosperms	Explain Salient features and Economic importance of Gymnosperms with examples	L	CO1	Understand	2.5
General account on Gymnosperms	List major timber plants in gymnosperms	L	CO1	Understand	0.5
General account on Gymnosperms	Differentiate Cycadales and Coniferales	L	CO1	Understand	2
Module II					
Vegetative parts of typical flowering plant	Define stem and its modification	L	CO2	Understand	1
Vegetative parts of typical flowering plant	Illustrate Root morphology with special emphasis on types (tap root, fibrous root, adventitious root) and modification of root (storage, mechanical support, respiration)	L	CO2	Understand	2
Vegetative parts of typical flowering plant	Explain parts of leaf, types of leaf (simple and compound leaf), venation and phyllotaxy	L	CO2	Understand	2
Flower morphology	Describe floral whorls, types of flower, fruit, placentation, pollination and fertilization	L	CO2	Understand	3
Inflorescence	Explain types of inflorescence (Racemose, Cymose and Special type) with examples	L	CO2	Understand	2
Angiosperm taxonomy	Define Binomial nomenclature	L	CO2	Understand	1

Angiosperm taxonomy	Describe identifying features of following angiosperm families with special emphasis on scientific names of economically important plants: 1. Annonaceae - Annona, Polyalthia 2. Fabaceae - Tamarindus, Accasia 3. Rubiaceae - Ixora, Hamelia 4. Verbenaceae - Lantana, Tectona 5. Poaceae - Paddy, Wheat 6. Malvaceae - Gossypium, Hibiscus 7. Euphorbiaceae- Rubber, Euphorbia 8. Myrtaceae - Eucalyptus, Syzigium	L	CO2	Understand	3
Angiosperm taxonomy	Define Herbarium and herbarium techniques	L	CO2	Understand	1
Module III					
Plant cell and cellwall	Explain ultrastructure and living and non living inclusions of cell	L	CO3	Understand	1
Plant cell and cellwall	Give an account on structure of cellwall	L	CO3	Understand	1
Plant cell and cellwall	Define pits and its types and plasmodesmata	L	CO3	Understand	1
Plant tissues	Differentiate Meristematic and Permanent tissues	L	CO3	Understand	1
Plant tissues	Define Meristem, its characteristics features and types of meristem	L	CO3	Understand	2
Plant tissues	Describe Permanent tissues and its types (simple and complex)	L	CO3	Understand	3
Theories of meristem	Describe structure of shoot apex and root apex	L	CO3	Understand	2
Theories of meristem	Explain theories of shoot apical meristem and root apical meristem	L	CO3	Understand	2
Vascular bundle	Differentiate different types of vascular bundles - conjoint, radial, collateral - open and closed - eccentric, concentric- amphivasal, amphicribal	L	CO3	Understand	2
Module IV					
Primary structure of stem, root and leaf	Differentiate primary structure of Dicot stem (Eupatorium)and Monocot stem (Grass)	L	CO4	Understand	2
Primary structure of stem, root and leaf	Differentiate primary structure of Dicot root (Tinospora)and Monocot root (Colocasia)	L	CO4	Understand	1.5
Primary structure of stem, root and leaf	Differentiate primary structure of Dicot leaf (Ixora) and Monocot leaf (Grass)	L	CO4	Understand	1.5
Secondary structure of stem and root	Explain normal secondary growth in dicot stem with special emphasize on cambial ring, cambial activity, secondary xylem and secondary phloem	L	CO4	Understand	2
Secondary structure of stem and root	Describe periderm formation - phellem, phellogen and phelloderm	L	CO4	Understand	2
Wood anatomy	Differentiate soft wood and hard wood	L	CO4	Understand	1
Wood anatomy	Define growth rings and Dendrochronology	L	CO4	Understand	0.5
Wood anatomy	Differentiate heart wood and sap wood	L	CO4	Understand	1
Wood anatomy	Explain structure of tyloses	L	CO4	Understand	1
Wood anatomy	Describe functions of lenticels	L	CO4	Understand	0.5
Wood defects	Explain different types of wood defects - Knots, shakes, grain issues, warping, insect damage	L	CO4	Understand	2

Suggested Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		
1	Prepare notes on plant diversity and classify plants into major groups Draw a flowchart showing the classification of plants Prepare self assessment questions on major plant groups	6
2	Create a glossary on botanical terms with meaning Read textbooks or online sections on fungi and wood defects caused by fungi Prepare short and long answer type questions on algae	6

3	Conduct group discussion on characteristic features of bryophyte, pteridophyte and gymnosperm Prepare a comparison table between Cycadales and Coniferales List out major timber plants in gymnosperms and their economic importance	6
4	Revise major topics in plant diversity Oral or written presentation of characteristic features of different plant groups	6
Module II		
5	Explore educational videos or use textbooks for reference Prepare a chart showing parts of a flowering plant	6
6	Conduct group discussion on morphology of flowering plants and modification Prepare a short report on root type, stem type, flower type and leaf arrangement of a selected flowering plant from your surrounding	6
7	Conduct quiz based on morphology and taxonomic terms Prepare short notes on taxonomic families with examples	6
8	Study local plants and note their common and scientific names Prepare a mini herbarium sheet of some common flowering plants	6
Module III		
9	Pre-reading and concept exploration Explore educational videos and animations on structure of cell	6
10	Prepare chart on types of plant tissues and their functions Practice diagrams of different plant tissues	6
11	Watch educational videos or animations on plant tissues Observe plant growth and identify the role of different meristems in plant growth Create a comparison chart detailing different theories of apical meristem	6
12	Note important terms and definition on plant anatomy Practice diagrams of different arrangements of vascular bundle	6
Module IV		
13	Prepare labelled diagram of dicot and monocot anatomy Watch an introductory video or read a textbook chapter on wood structure	6
14	Prepare a comparison table of heartwood and sapwood, hard wood and soft wood Create a glossary on wood related terms with meaning	6
15	Prepare a wood defect identification chart Conduct group discussion on major topics of wood anatomy and wood defects	6
Total Self-Learning hours		90

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Botany for Degree students Fungi	B.R.Vashishta	S Chand publication, 2014.
2	Botany for Degree students, Algae	B.R.Vashishta	S Chand publication, 2014.
3	Botany for Degree students, Bryophyte	B.R.Vashishta	S Chand publication, 2014.
4	Plant anatomy	B.P.Pandey	Chand Publishing, 2012
5	Economic Botany	v	Chand Publication.1999
6	Taxonomy	Singh&Jain	Rastogi Publications

REFERENCE			
Sl. No.	Title	Author	Publication
1	College Botany Vol.I, II, III		New central books agency, 2011

WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	https://uou.ac.in/sites/default/files/slm/BOT(N)-101.pdf	I
2	https://www.mdpi.com/2079-6412/10/12/1210	I

3	https://agroswamp.com/wp-content/uploads/An-Introduction-to-Plant-Taxonomy.pdf	II
4	https://www.biologyonline.com/tutorials/plant-tissues	III
5	https://buat.edu.in/wp-content/uploads/2023/08/Wood-Anatomy-Manual-final-PMEC-N1.pdf	IV
6	https://www.slideshare.net/slideshow/defects-in-timber/14918367	IV

Assessment Methodology - Theory

Mode	Attendance	CA1	CA2	CA3	CA4	CA5	CA6	ESE
		Self-learning activities (Module 1)	Self-learning activities (Module 2)	Self-learning activities (Module 3)	Self-learning activities (Module 4)	Series Exam (2 modules)	Series Exam (2 modules)	Written Exam (theory)
Duration						2 hours	2 hours	2.5 hours
Exam mark		15	15	15	15	50	50	60
Converted to						20	20	
Marks	5	15				20		60
Total	40							60
Tentative schedule	End of semester	By 4th week	By 7th week	By 11th week	By 14th week	8th week	15th week	End of semester

Series Examination Pattern- Theory

Time	Module	Part A	Marks	Part B	Marks	Part C (With choice)	Marks	Total
2 hours	Any one module	2	1	3	3	2	7	25
	Any one module	2	1	3	3	2	7	25
	Number	4		6		4		50

End Semester Examination Pattern of Question Paper - Theory

Duration	Module	TYPE OF QUESTIONS						Total Marks
		Part A (2 Questions from each module)		Part B 2 out of 3 questions from each module)		Part C (1 set questions with choice from each module)		
		No. of Questions	Marks (1 mark each)	No. of Questions	Marks (3 marks each)	No. of Questions	Marks (7 marks each)	
2.5 hours	1	2	2	3	6	1 set	7	15
	2	2	2	3	6	1 set	7	15
	3	2	2	3	6	1 set	7	15
	4	2	2	3	6	1 set	7	15
		8	8	12	24	4 sets	28	60